

AllSpeak — programming in any language

A working implementation of multilingualism at the level of the code itself

Graham Trott, independent developer · info@allspeak.ai · <https://allspeak.ai>

The gap

UNESCO's multilingualism agenda has addressed how people read and write prose: digital content, interfaces, translation, domain names and email addresses, and the revitalisation of Indigenous languages. One layer is untouched — **the language in which people read and write logic**. Every mainstream programming language uses English keywords, and code is not text to be translated but structure to be understood.

This matters more, not less, as AI writes more code. AI has no difficulty writing in a small, constrained grammar, and the population that must *read, check and maintain* the result is no longer limited to English-literate professionals. Comprehension, not authorship, is the binding constraint.

What AllSpeak is

An open-source scripting runtime in which the code itself is written in the learner's language. A single language-neutral engine executes every variant; language packs supply the keyword vocabulary and grammar mappings. Most localisation stops at the documentation; AllSpeak stops at the keywords.

Keywords such as `put`, `create`, `append`, `rest get` and `on click` conceal HTTP requests, DOM manipulation and data flow behind words anyone can read — over an engine that supports 127 opcodes, concurrency, a JSON pipeline, graphics, maps and messaging.

Evidence

- **Four language packs operational** — English, French, German, Italian. The non-English packs are functional and awaiting native-speaker educator validation; the architecture admits any language, including non-Latin scripts.
- **A deployed business system** — approximately 4,500 lines across five modules, in daily use for bookings, expenses and financial records.
- **A 20-step tutorial** in all four languages, and a white paper with DOI (10.5281/zenodo.22018537) describing the architecture and the method for adding a language.
- **Built by one person**, with AI assistance, and open source.

Points of contact

- **2003 Recommendation** — language technologies, multilingual digital content, equitable access. AllSpeak is a language technology at the syntactic level.
- **Internet Universality ROAM** — Accessibility and Multi-stakeholder participation. Any language community can extend the runtime, and the method is documented.
- **Universal Acceptance for a multilingual Internet** — the same argument one layer deeper: not only domain names and addresses, but source code.
- **Global Roadmap on Multilingualism in the Digital Era** — a working answer to the call for industry to “design inclusive Language Technologies”.

The question

Could a project of this kind contribute usefully to UNESCO's work — as a case study, a reference implementation, or a contribution to its consultations — and if so, through which route? The project is unfunded and seeks collaborators rather than funding.

Correspondence: info@allspeak.ai